

RESEARCH INTERESTS

Quantum many-body systems, condensed matter theory, quantum information theory, mathematical physics, theoretical and scientific machine learning.

EDUCATION

- **University of Lisbon - Instituto Superior Técnico (IST)** Lisbon, Portugal
MSc in Applied Mathematics and Computation - Pure Mathematics Track (GPA: 17/20) 2025
Thesis: Semilinear Wave Equations on Decelerated Expanding FLRW Spacetimes (Grade: 19/20)
- **École Polytechnique Fédérale de Lausanne (EPFL)** Lausanne, Switzerland
Swiss-European Mobility Programme in Theoretical Physics 2017
- **University of Lisbon - Instituto Superior Técnico (IST)** Lisbon, Portugal
BSc in Engineering Physics - Theoretical Physics Track (GPA: 15/20) 2019

ACADEMIC POSITIONS

- **Invited Assistant Professor at ISCTE – Instituto Universitário de Lisboa** Lisbon, Portugal
Taught Calculus I, Calculus II and Linear Algebra to 1st year Economics students. 2024-2025
- **Teaching Assistant at ISCTE – Instituto Universitário de Lisboa** Lisbon, Portugal
Review office hours in Calculus I for 1st year international Management students. 2024
- **Research Assistant at Aalto University (ASci Program)** Helsinki, Finland
Symmetry and equivariance properties of Graph Neural Networks under the supervision of Prof. Vikas Garg. 2023

WORK EXPERIENCE

- **Private Tutoring in Physics and Mathematics** Lisbon, Portugal
Tutored several students in introductory math and physics courses at the high-school and university level. 2014 - 2023
- **Translation at Altice Foundation** Lisbon, Portugal
Translation of mathematical content on Khan Academy from English to Portuguese. 2019-2020
- **Summer Research Intern at Instituto Gulbenkian de Ciência (IGC)** Oeiras, Portugal
Modelled the evolution of cancerous cells under the supervision of Dr. Mónica Dias and Dr. Claudia Bank. 2016

SUMMER SCHOOLS

- **Quantum Landscapes International School** Paris, France
Advanced lectures on quantum many-body systems, tensor networks, and quantum information. 2026
- **LxMLS22 - Lisbon Machine Learning Summer School** Lisbon, Portugal
Coursework in deep learning, probabilistic models, reinforcement learning, and natural language processing. 2022

SELECTED ACADEMIC PROJECTS

- **Mathematical Foundations of Machine Learning:** Wrote a set of notes exploring the basics of Learning Theory, Machine Learning and Kernel Methods, see [mathematical foundations of deep learning].
- **Manifold Learning:** Two projects for the courses in mathematics for machine learning and nonlinear optimization on manifold learning, see [manifold learning].
- **Feature Selection:** A project for the course in mathematics for machine learning on feature selection via information theoretic considerations, see [feature selection].
- **Collaborative Filtering:** A project for the course in mathematics for machine learning on collaborative filtering for a movie recommendation system based on matrix factorization methods, see [collaborative filtering].
- **Geometric Quantization:** A project exploring the basics of symplectic geometry and geometric quantization, see [geometric quantization].

SKILLS SUMMARY

- **Languages:** Portuguese (Native); English (Fluent).
- **Programming Languages:** Python (Pytorch/Numpy/Scikit-learn/Pandas); C/C++; R.
- **Tools:** Mathematica; L^AT_EX.